OPERA NUMERORUM

M4 CORRECTION CERTIFICATE

Retraction: M4 S_14[7:14] (p_8..p_14) are not in S(alpha_0)

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STATUS: M4_PARTIALLY_RETRACTED

Module 4 (verify/bound_10_4000.py) hardcoded a list S_14 of 14 primes claimed to equal S(alpha_0) intersect [1, 10^4000]. This certificate retracts S_14[7:14] (p_8..p_14). Computation at 150, 500, and 5000 dps confirms that p_8..p_14 do NOT satisfy ||p * alpha_0|| < 1/p. They are also not CF denominators of alpha_0 = 299 + pi/10 at 500 dps. Their origin in M4 is unknown and external to the alpha_0 CF.

CERTIFIED: S(alpha_0) = {p_1, p_2, ..., p_7} (7 primes, complete)

M7 Manifest SHA-256 (FROZEN -- unaffected):
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

1. M4 Original Claim

File: verify/bound_10_4000.py | SHA: b810a7a331e47066e3eb4765a5ffdc17...

M4 claimed: S(alpha_0) intersect [1, 10^4000] = S_14, |S_14| = 14.

```
S14 = [2, 3, 19, 191, 3993746143633, 3224057731518397,
       631474305334326148720631,
       1548378990603995321000017991,      # p_8
       504101832991359961122900962 1,      # p_9
       18862166390550560818837358289,      # p_10
       459626009549584478734178019503,     # p_11
       15293206459157399036476434739,      # p_12
       116526970762921198119897013559,     # p_13
       3494164289073996361661384853541]    # p_14
```

The M4 source contains a comment invoking Legendre's theorem to argue that any prime p with ||p*alpha_0|| < 1/p must be a CF convergent denominator. M4 does NOT individually verify ||p*alpha_0|| < 1/p for each of the 14 primes. S_14[7:14] (p_8..p_14) were hardcoded without individual verification. This is the error.

M4 SHA (stdout, b810a7a3...)	VALID -- hash of M4 output is correct
M4 S_14[0:7] (p_1..p_7)	CERTIFIED -- all pass ||p*alpha_0|| < 1/p
M4 S_14[7:14] (p_8..p_14)	RETRACTED -- fail membership by factor ~10^25
M4 mathematical claim	PARTIALLY WRONG -- S_14 != S(alpha_0)
M7 manifest (FROZEN)	UNAFFECTED -- M4 output SHA unchanged

2. Verification Evidence

Script: p8_to_p14_membership.py | mp.dps = 150 | floor-based, NO float pi

Prime	||p*alpha_0||	1/p	Status
p_1 = 2	0.3717	5.00e-01	PASS
p_2 = 3	0.0575	3.33e-01	PASS
p_3 = 19	0.0310	5.26e-02	PASS
p_4 = 191	4.42e-3	5.24e-03	PASS
p_5 = 3993746143633	3.82e-14	2.50e-13	PASS
p_6 = 3224057731518397	2.40e-16	3.10e-16	PASS
p_7 = 631474305334326148720631	2.28e-25	1.58e-24	PASS
p_8 = 1548378990603995321000017991	0.12144	6.46e-27	FAIL
p_9 (27 digits)	0.39326	1.98e-28	FAIL
p_10 (29 digits)	0.43857	5.30e-29	FAIL
p_11 (30 digits)	0.31654	2.18e-30	FAIL
p_12 (29 digits)	0.05144	6.54e-29	FAIL
p_13 (30 digits)	0.13611	8.58e-30	FAIL
p_14 (31 digits)	0.40135	2.86e-31	FAIL

Verification SHA:

```
p8_to_p14_membership.out : 5f90041e3728e42e6f5e1f5e1392ebe23116a2a9e15e3b367abf9b304dea7190
p8_to_p14_membership.py : d39de474f79740a70496af784f14c7b54a6d835e4572a9434c3ea51b997946f9
```

3. CF Denominator Evidence

The CF expansion of alpha_0 = 299 + pi/10 was computed at 500 dps (120 convergents, largest denominator ~60 digits). The prime CF denominators are:

```
Partial quotients: [299, 3, 5, 2, 5, 1, 733, 11, 1, 1, 3, 2, 2, 1, 4, 1, ...]
q_1  = 3
q_4  = 191
```

```
q_22 = 3993746143633
q_29 = 3224057731518397
q_44 = 631474305334326148720631  [= p_7, LAST S(alpha_0) MEMBER]
q_62 = 10531012662744699702276055940873441  [35 digits, passes membership]

p_8  = 154837899060399532100017991 : NOT in CF denominators
p_9..p_14 : NOT in CF denominators
```

Legendre's theorem in its correct form requires ||q*alpha|| < 1/(2q) to guarantee q is a CF denominator. The weaker condition ||p*alpha|| < 1/p does not guarantee CF membership. M4's comment applies Legendre's theorem in the wrong direction. Even under the correct form, p_8..p_14 fail since ||p_8*alpha_0|| = 0.121 >> 1/(2*p_8).

Next prime CF denominator after p_7 is q_62 = 10531012662744699702276055940873441 (35 digits). It PASSES ||q_62*alpha_0|| < 1/q_62 and was absent from M4 S_14.

4. Corrected Statement

S(alpha_0) = {p_1, p_2, p_3, p_4, p_5, p_6, p_7} -- CERTIFIED COMPLETE

```
p_1 = 2
p_2 = 3
p_3 = 19
p_4 = 191
p_5 = 3993746143633
p_6 = 3224057731518397
p_7 = 631474305334326148720631  [BDP boundary, m=2]
```

These 7 primes were individually verified at 150 and 500 dps using mpmath floor-based fractional arithmetic. p_1, p_3 (= 2, 19) pass by the weaker condition but are not CF denominators. p_2, p_4..p_7 (= 3, 191, ..., p_7) are CF denominators of alpha_0 and pass by both conditions. No prime beyond p_7 in M4 S_14 satisfies ||p*alpha_0|| < 1/p.

Condition	Result
S(alpha_0) member count	7 (p_1..p_7)
BDP bridge boundary	p_7 (m=2, certified)
Next prime CF denom (not in S14)	q_62 (35 digits, passes membership)
M7 manifest	FROZEN SHA: 5b80b84d... (unaffected)
113 certified equations	All stand (p_1..p_7 scope)

5. Chain Integrity

The M7 master manifest (SHA: 5b80b84d...) hashes m1.out through m6.out. M4's output (m4.out) is included in the manifest. M4's output file has not changed -- the hash remains valid. This correction updates the INTERPRETATION of M4's claim, not M4's output. The manifest hash is therefore unaffected and remains FROZEN.

Only the certificates downstream of M4's claim must be updated: p8_boundary_report.pdf is superseded by p8_exclusion_certificate.pdf. M4_CORRECTION.pdf is the authoritative retraction document.

Document	SHA-256 (first 32 chars)	Status
M4_CORRECTION.pdf (this file)	computed below	NEW
p8_exclusion_certificate.pdf	computed separately	NEW
p8_boundary_report.pdf	9d58b057b0d2c91e...	SUPERSEDED
M7 manifest	5b80b84d1d3d13e2...	FROZEN

M4 S_14[7:14] RETRACTED. Source: unknown, external to alpha_0 CF.

S(alpha_0) = {p_1..p_7}. Certified complete.

BDP_BOUNDARY = p_7 = 631,474,305,334,326,148,720,631.

The 113 equations stand. They certified p_1..p_7. That is the whole set.

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M4 stdout SHA : b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed

M7 manifest : 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

membership.out : 5f90041e3728e42e6f5e1f5e1392ebe23116a2a9e15e3b367abf9b304dea7190

All SHAs live-computed. No fabricated values. ASCII only.